

CHES 2026 TUTORIAL · ANTALYA, TÜRKİYE

Quantum Physical Unclonable Functions: Taxonomy, Implementations, and Open Problems: Lab Session

Dr. Mahshid Delavar

Lecturer (Assistant Professor) in Cybersecurity
University of Sheffield, UK

Dr. Meltem Kurt Pehlivanoğlu

Lecturer (Associate Professor), Computer
Engineering
Kocaeli University, Türkiye

Oğuz Narlı

Senior Embedded Software Engineer
CTech Information Technologies, Türkiye

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Getting the environment ready

- 01 Clone the repo** (notebooks + Pictures/ folder)

```
git clone https://github.com/oguznrl/QPUF-Tutorial
```

- 02 Create a virtual environment** with Python 3.10–3.12 (Windows: qpuf-env\Scripts\Activate.ps1)

```
python3 -m venv qpuf-env && source qpuf-env/bin/activate
```

- 03 Install everything listed in requirements.txt** (TensorFlow, Qiskit, Jupyter)

```
pip install -r requirements.txt
```

- 04 Install pypuf without its dependencies:** pypuf 3.2.1 pins TensorFlow 2.4, which does not exist for Python 3.9+

```
pip install --no-deps pypuf==3.2.1
```

- 05 Open the notebooks** in Jupyter or VS Code and select the **qpuf-env** kernel. On Google Colab, skip the venv and run steps 3–4 in a cell with %pip.

- 06 TypeError: Expected int8?** Run the patch cell after the attack, then re-run the attack.



Scan for the repo

[github.com/ oguznrl/
QPUF-Tutorial](https://github.com/oguznrl/QPUF-Tutorial)

Cloning a classical PUF in software

Physically unclonable, but mathematically modelable.

01

Goal

- Simulate a 64-bit, $k=5$ XOR-APUF with pypuf
- Measure quality metrics: reliability, uniqueness, uniformity
- Run an LRAttack2021 modeling attack on 200,000 CRPs

02

What we want to see

- Additive delay model: each component is linear in $\varphi(c)$
- XOR breaks linearity, but does not remove learnability
- Good quality metrics do not mean resistance to ML attacks

03

Expected result

- Uniqueness ≈ 1 , uniformity ≈ 0.5
- Reliability < 1 : noise accumulates through the XOR
- High attack accuracy: **$> \sim 90\%$** $\rightarrow$ a working software clone

Hiding the response inside a quantum state

How much harder does the same attack get when the response arrives as a qubit?

01

Goal

- Encode the same CPUF's response from Lab 1 into a qubit via BB84
- Derive the basis bit from a second, independent PUF
- Run the same LRAttack2021 against a single-copy Eve

02

What we want to see

- No-cloning and non-orthogonal states: Eve cannot learn the basis
- Even the optimal measurement (Breidbart, Helstrom bound) leaves ~14.6% label noise
- The analytical shortcut and the Qiskit circuit give the same result

03

Expected result

- $p_{\text{guess}} \approx 0.8536$, $p_{\text{error}} \approx 0.1464$ with both methods
- No accuracy ceiling: required CRPs $\approx 1/(1-2p)^2 = 2\times$
- Attack accuracy **~50%**, vs. **>~90%** in Lab 1